KAY HWANG

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SUMMARY

A data science and machine learning student with strong Python programming skills and experience in data analysis, scientific/statistical computing, and SQL-based data analysis. Experience in building analytical tools that rely on programming and statistical methods.

SKILLS

- Programming: Python, SQL, Java, C++
- Data Science: Statistical Analysis, Data Cleaning/Visualization, MLM development
- Libraries/ Tools: NumPy, SciPy, Pandas, Mathplotlib, Scikit-learn, Tensorflow, Git

CERTIFICATIONS AND AWARDS

AP Scholar: AP Statistics(5), AP Calculus AB(5), AP Computer Science A(5), AP Biology(5), Johns Hopkins University: Intro to Genomic Technologies, Johns Hopkins University: Python for Genomic Data Science, Stanford Online: Supervised Machine Learning: Regression and Classification, Stanford Online: Advanced Learning Algorithms

WORK EXPERIENCE

OPENBIO @ HARVARD
Cambridge, Massachusetts

Research Team Member 10/2025 to Current

- 12/150 selected
- Leading and supporting student-led research, through development of NumPy/Scipy based data match algorithm
- Collaborating with peers to develop and refine low-cost, data-driven tools for education and innovation in the life sciences.

CHILDREN'S SPECIALIZED HOSPITAL - RUCARES
New Brunswick, NJ

Research Intern 06/2024 to 06/2025

- Create Python-based programs to automate data cleaning and visualization models
- Used Python, SQL and Javascript to code tools for data collection.
- Conduct research on Extinction Theory w/ Dr. Wayne Fisher, applying statistical modeling and data analysis

THE TECHTRUM ALLIANCE

Founder/CEO 03/2024 to Current

- Created and registered a 501(c)3 certified non-profit organization with 7 sectors
  Lead members age 16-57 from 3 continents; partnered w/ autism advocacy org. raising $5k+ for iPads for Autistic people
- Create videos on tech-literacy, safe tech-usage, and the importance of technology for Autistic Individuals

PROJECTS

NBBO Analysis + Prediction (https://github.com/KayHwang0529/NBBO-Analysis)

- Two-part stock price analysis and prediction tool applied to a dataset on past NIFTY 50 metrics
  - Predicted closing price with RMSE = 28.375, MAE=21.556, MAPE=1.89%
- Python, Seaborn, NumPy, Pandas, Tensorflow, Matplotlib, SciPy

Autism Screening Model (https://github.com/KayHwang0529/Autism-Research)

- Neural network used to classify whether an individual is likely to have Autism Spectrum Disorder based on behavioral/phenotypic features
  - Predicted Autism diagnosis with MAPE= 4.35%
- Python, Seaborn, NumPy, Pandas, Tensorflow, Matplotlib, SciPy

Task Manager Web App (https://github.com/KayHwang0529/InventoryApp)

- Track Homework assignments and dynamically update UI to reflect completion status
- SQL, Javascript, HTML CSS

EDUCATION

B.A.: STATISTICS AND CS
Harvard University, Cambridge, MA
Expected Graduation 05/2029

HIGH SCHOOL DIPLOMA
Bergen County Academies, Hackensack, NJ

- Graduation 05/2025
- GPA: 3.97/4.0 (unweighted)

Relevant Course Work

- SQL
- Data Driven AI, AI
- CS50
- AP CSA
- Intro to Web App Design